Build Your Own AI Coding Agent with GitHub Copilot

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AI-native thinking

- Build a minimal AI coding agent using GitHub Copilot.
- Learn how modern coding agents understand repositories, plan changes, edit code, and verify their work.
- Develop practical AI software engineering skills for collaborating effectively with coding agents.
- USE AI TO UNDERSTAND AND BUILD AI

Introduction

- Three ways you can run an LLM
- Run it locally on your laptop
- Run it on the cloud by calling an API from a service provider
- Run it indirectly by using LLM-backed product (e.g. GitHub Copilot, ChatGPT, ...)

Outline

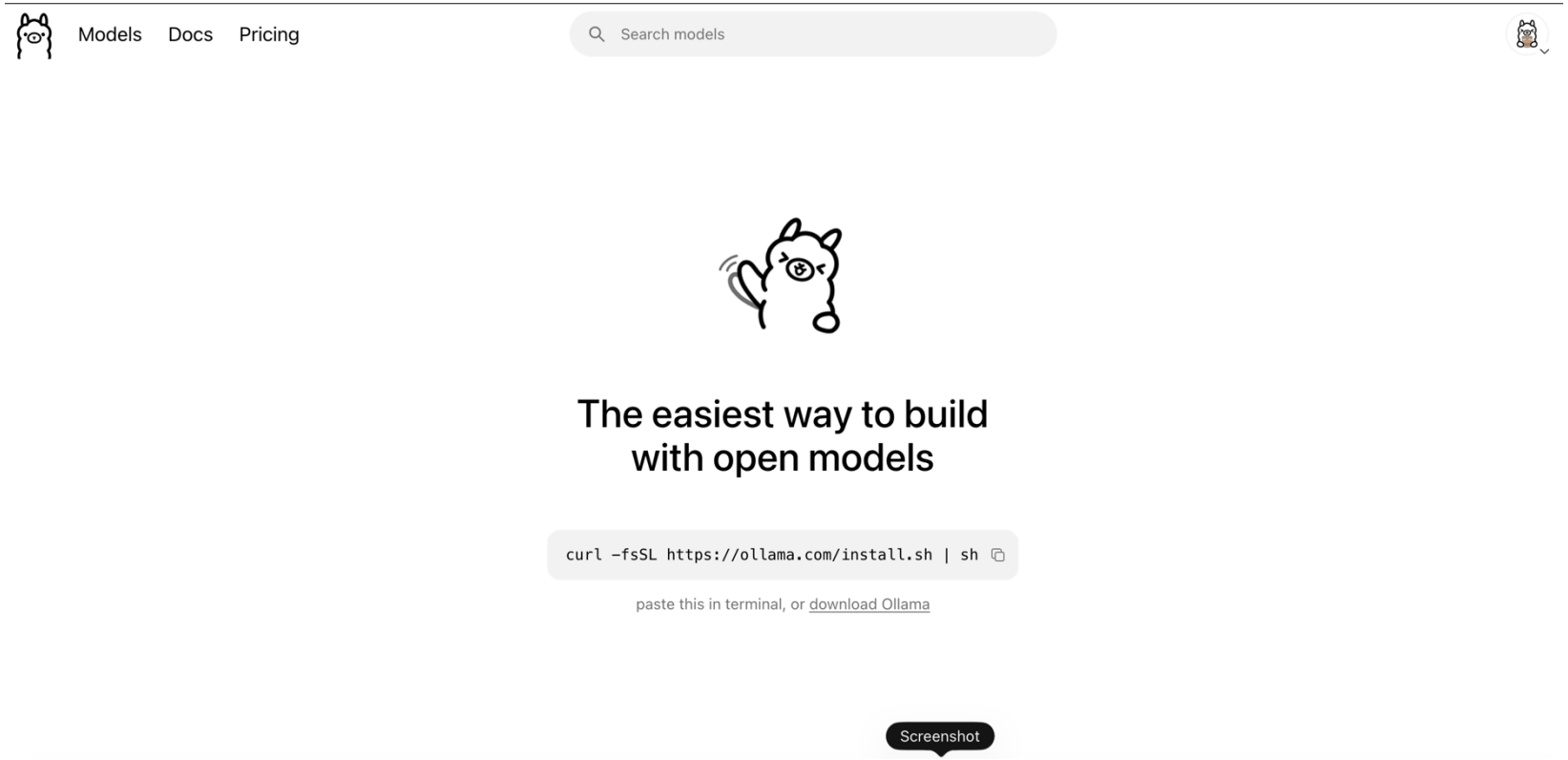
- STEP01: running a local LLM on your laptop
- STEP02: calling a cloud LLM through API
- STEP03: build a mini coding agent with a cloud LLM

Prerequisite

- GitHub Copilot required for all three steps.
  - Codex or Claude Code also works
  - It's OK if you still do not have access to it yet. We will share the materials and recording with you after the session. You can just watch me demo and try on your own afterwards.
- VS Code and a working Python environment (Anaconda)
- STEP01: running a local LLM on your laptop
  - Need to download ollama, but you can skip this step. Local LLMs are pretty weak. They cannot handle complex task. I will show you.
- STEP02: calling a cloud LLM through API
  - Need to top up in OpenRouter. The minimum top up rate is 5 USD.
- STEP03: build a mini coding agent with a cloud LLM
  - Same as STEP02.

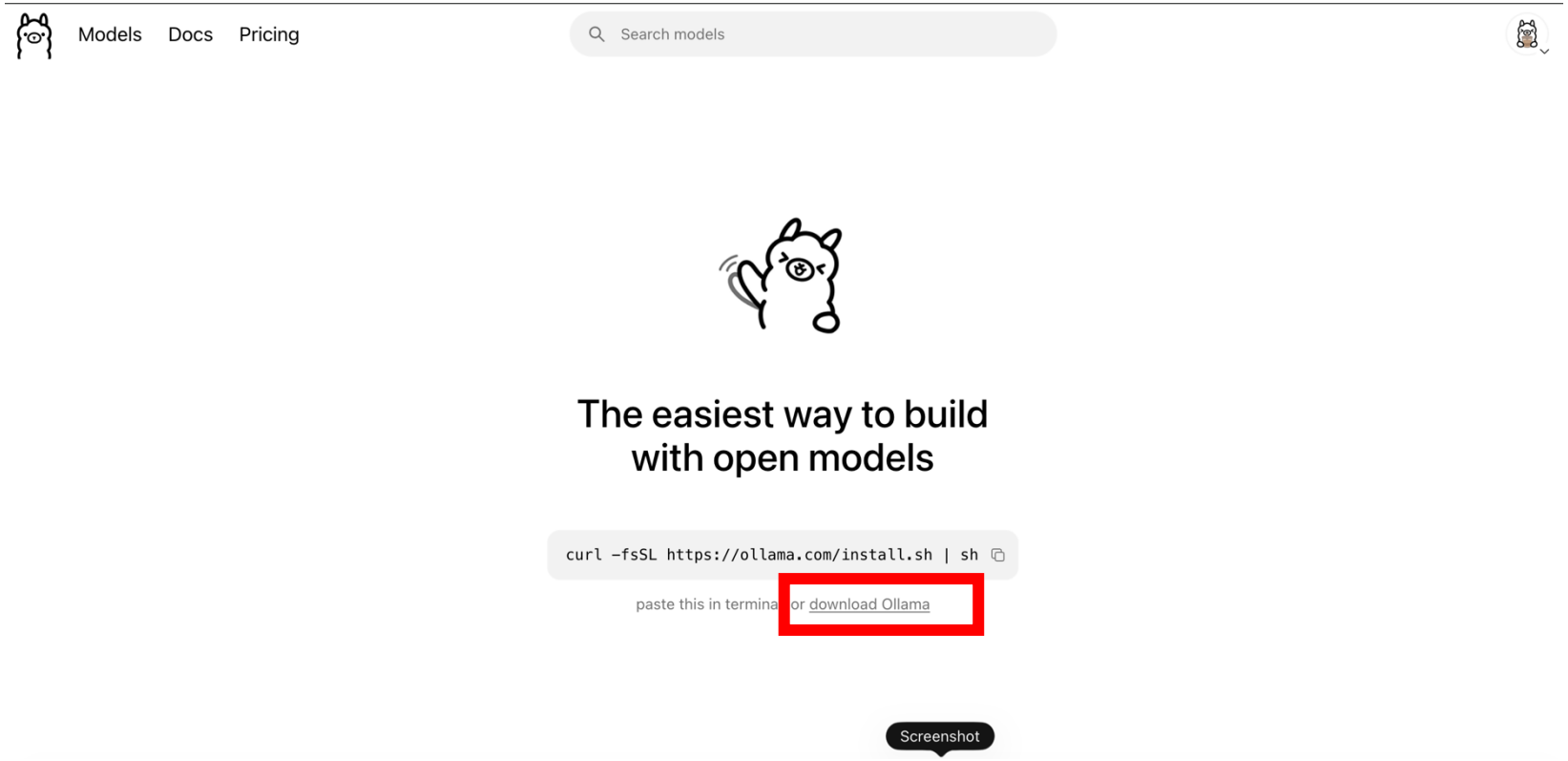
STEP01: running a local LLM on your laptop

- Download Ollama
- Search “Ollama”

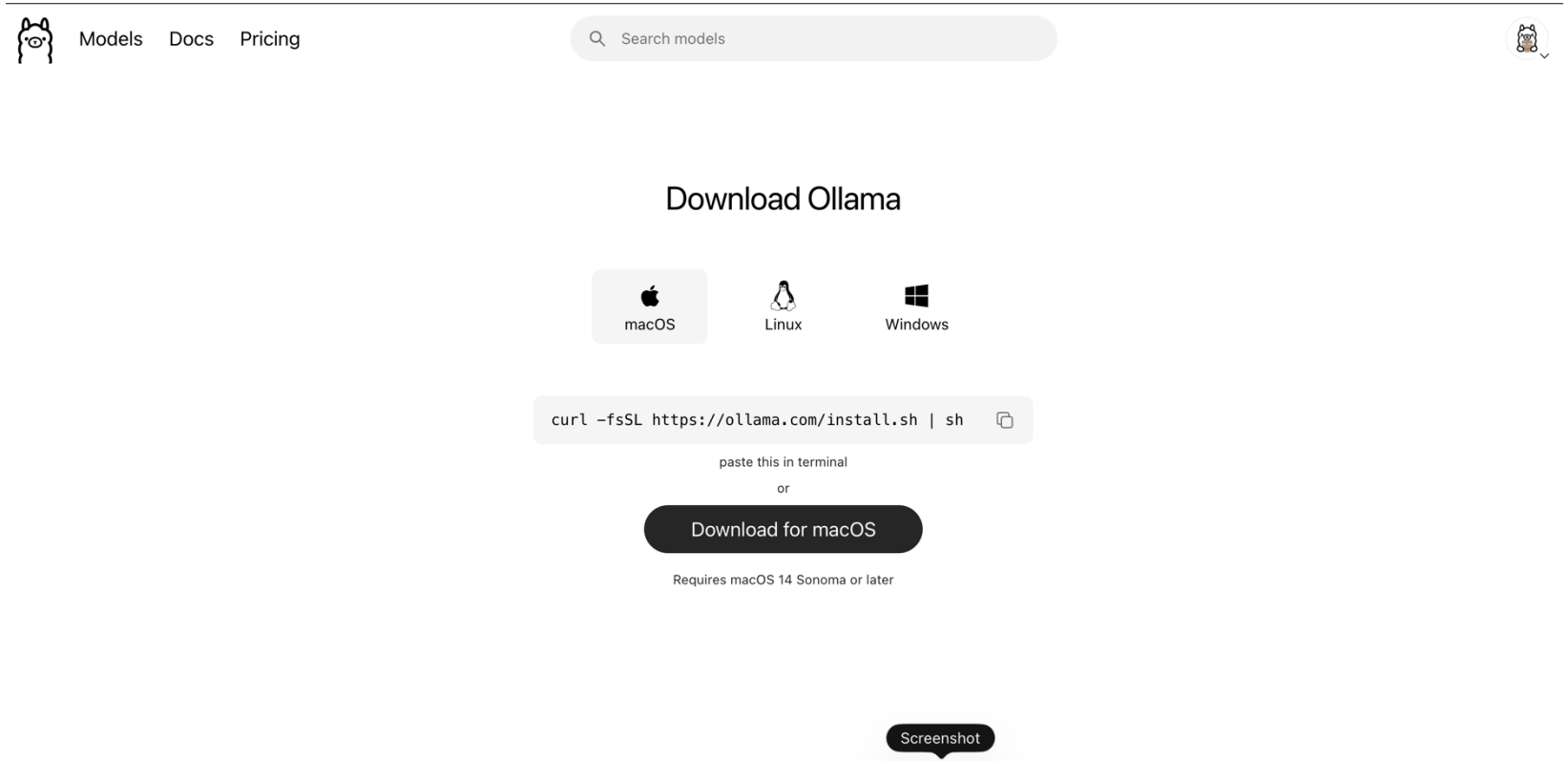


STEP01: running a local LLM on your laptop

- If you don’t know how to use the terminal,



STEP01: running a local LLM on your laptop



STEP01: running a local LLM on your laptop

- It takes a while to download and install.
- If it doesn’t work on your laptop, just watch me doing the demo.

STEP01: running a local LLM on your laptop

- Ask GitHub Copilot to act on SETUP.md
- It will help examine whether your ollama is properly installed or not.
- If ollama is properly installed, then it will further help you install the local LLM we need with ollama.
- It will also help setup the python environments for us.
- It takes a while.
- Please just allow the actions that GitHub Copilot is going to take.
- If it asks you to choose a Python environment (we assume you have set up earlier), choose the one with the latest Python version (if you have many)

STEP01: running a local LLM on your laptop

- We will use Qwen3 0.6B to run on local laptop.

Practical requirements

Configuration	Minimum	Recommended
System RAM	4 GB	8 GB or more
Free storage	1–2 GB	3–5 GB
Dedicated GPU	Not required	Optional
GPU VRAM	Not required	2 GB or more
CPU	Modern 64-bit dual-core	Recent 4-core or better
Operating system	Windows, macOS, or Linux	Any recent version

For comparison, Ollama's quantized `qwen3:0.6b` download is approximately **523 MB.** Ollama

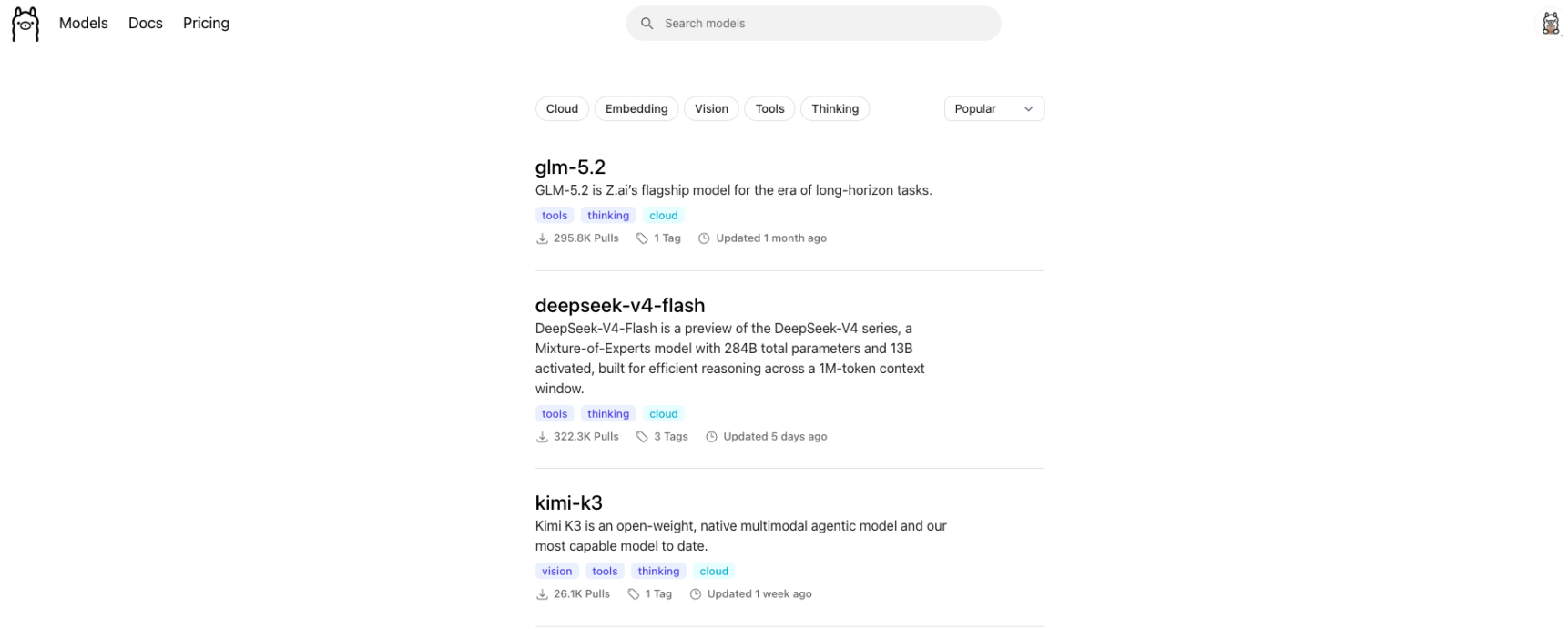
Estimated memory usage

The actual memory depends on the model format and context length.

Model format	Weight memory	Typical total RAM during use
FP32	About 2.4 GB	3.5–5 GB
FP16/BF16	About 1.2 GB	2–4 GB
INT8	About 0.6–0.8 GB	1.5–3 GB
INT4, such as Ollama/GGUF	Ab 0.5–0.7 GB	1–2.5 GB

- An **8 GB laptop** should run it easily.
- A **16 GB laptop** has ample headroom.
- A **4 GB laptop** may run the quantized version, but the operating system could create memory pressure.
- An integrated Intel, AMD, or Apple GPU is sufficient; CPU-only inference also works.

STEP01: running a local LLM on your laptop



STEP01: running a local LLM on your laptop

Current frontier-model sizes

Model	Architecture	Total parameters	Active per token	Publicly disclosed?
GPT-5.6 Sol	Undisclosed	Unknown	Unknown	No
Claude Fable 5	Undisclosed	Unknown	Unknown	No
Claude Mythos 5	Undisclosed	Unknown	Unknown	No
Kimi K3	MoE	2.8T	104B	Yes
DeepSeek V4 Pro	MoE	1.6T	49B	Yes
DeepSeek V4 Flash	MoE	284B	13B	Yes
Qwen3.8-Max	MoE	2.4T	Approximately 95B	Yes/reported
Qwen3-0.6B	Dense	0.6B	0.6B	Yes

>1000x

STEP01: running a local LLM on your laptop

- Task: we would like to ask Qwen3 0.6B the following question picked from Winograd Schema Challenge

```
WSC_QUESTION = (
    "Winograd Schema Challenge question "
    "(Hugging Face dataset: winograd_wsc, config wsc273):\n\n"
    "Sentence: The trophy doesn't fit into the brown suitcase "
    "because it is too big.\n\n"
    "Question: What is too big \u2014 the trophy or the suitcase?\n\n"
    "Give your answer and a one-sentence justification. You do "
    "not need to verify whether your answer matches the "
    "dataset's official label."
)
```

What should be the correct answer?

STEP01: running a local LLM on your laptop

- Task: we would like to ask Qwen3 0.6B the following question picked from Winograd Schema Challenge
- Winograd Schema Challenge proposes a dataset to ask computer (AI) to answer a certain type of questions regarding the semantic understanding of natural language.
- AI didn’t well in answering this kind of questions until we have LLMs.

```
WSC_QUESTION = (
    "Winograd Schema Challenge question "
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    "Give your answer and a one-sentence justification. You do "
    "not need to verify whether your answer matches the "
    "dataset's official label."
)
```

STEP01: running a local LLM on your laptop

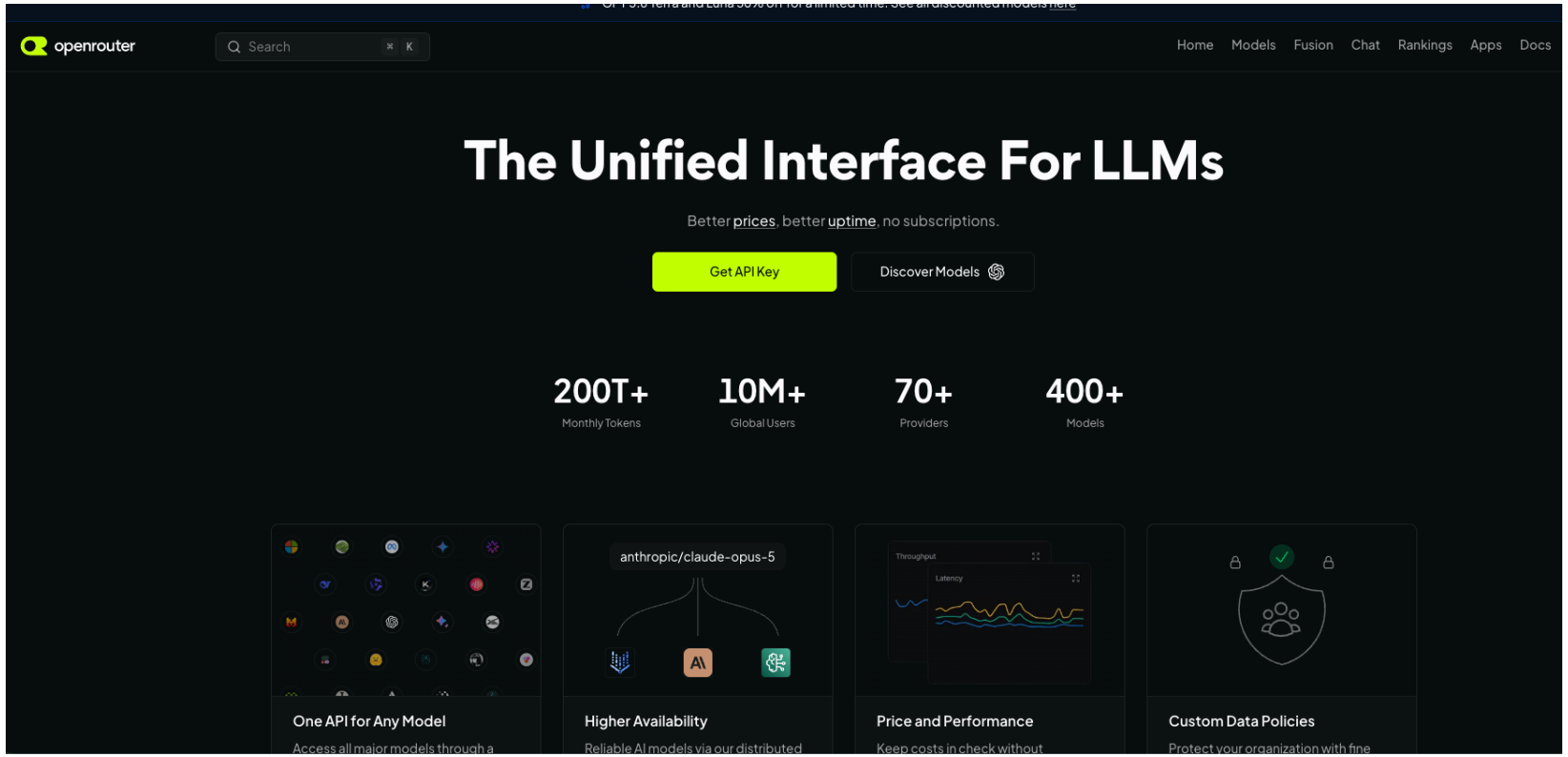
- Basic features of GitHub Copilot
- Ask: understand the code by clarifying with GitHub Copilot
- Plan: plan before complex tasks (you can refine the plan interactively until you deem appropriate to execute)
- Agent: “autopilot”
- And more….

STEP01: running a local LLM on your laptop

- What is the output of qwen3:0.6b?
- Can it produce the correct answer?
- We change to a larger model.

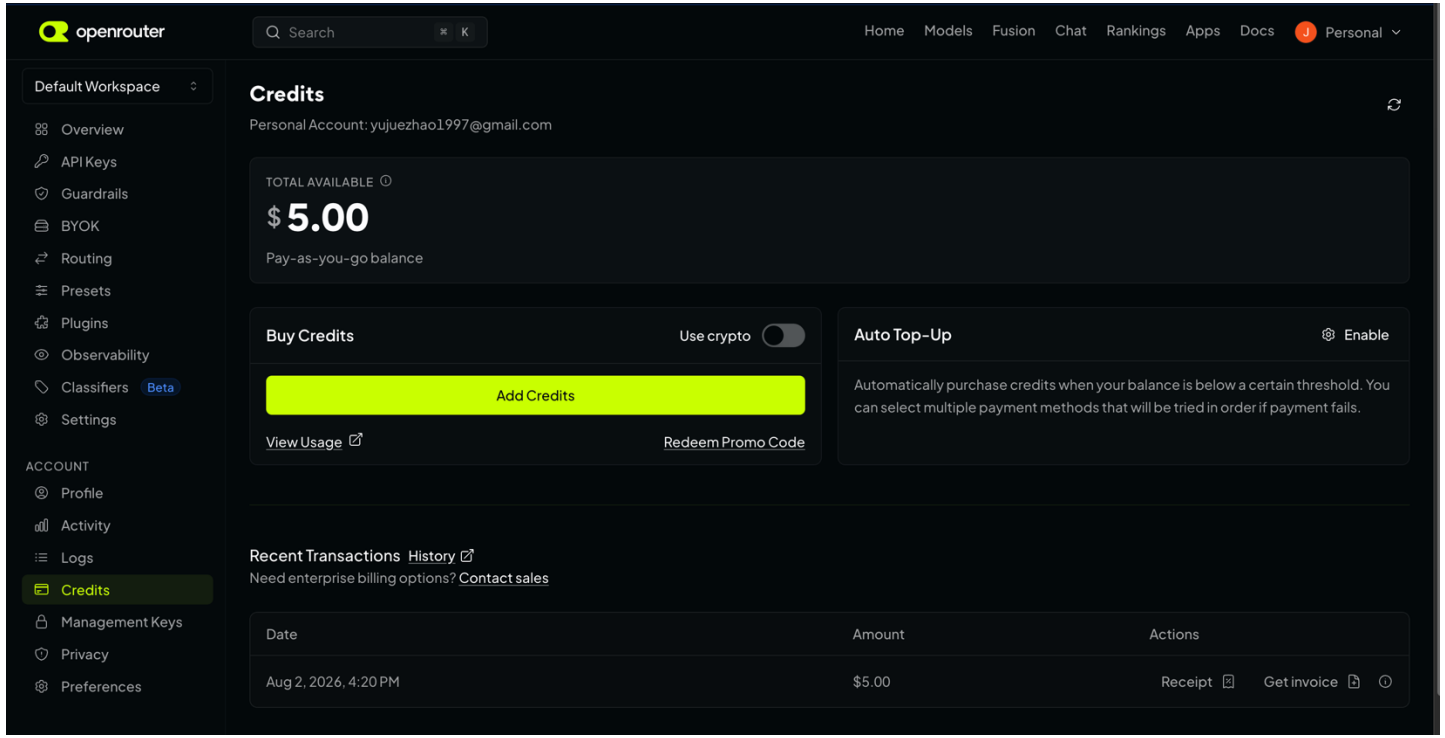
STEP02: calling a cloud LLM through API

- If you want to follow through, you will need to sign up on OpenRouter



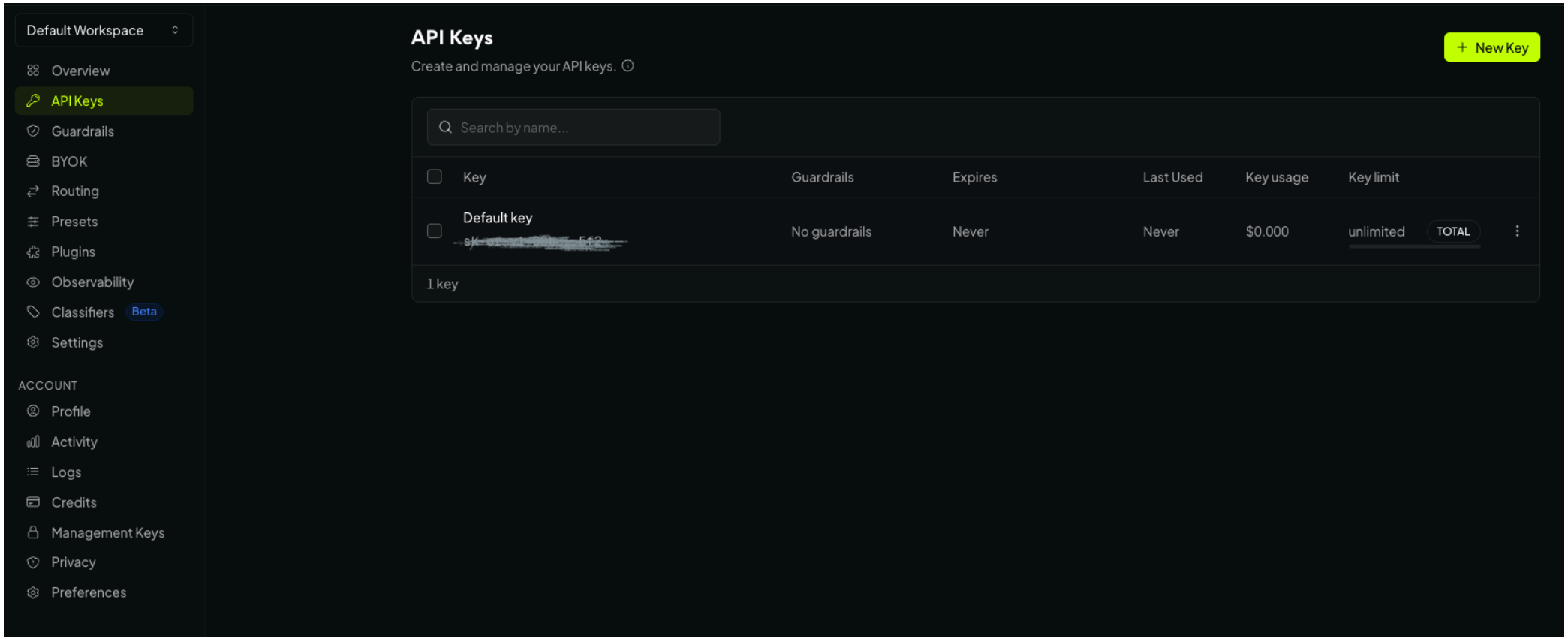
STEP02: calling a cloud LLM through API

- We will need to top up some $ if we want to call the frontier models. The min top up rate is 5 USD, which is more than enough for our mini project.



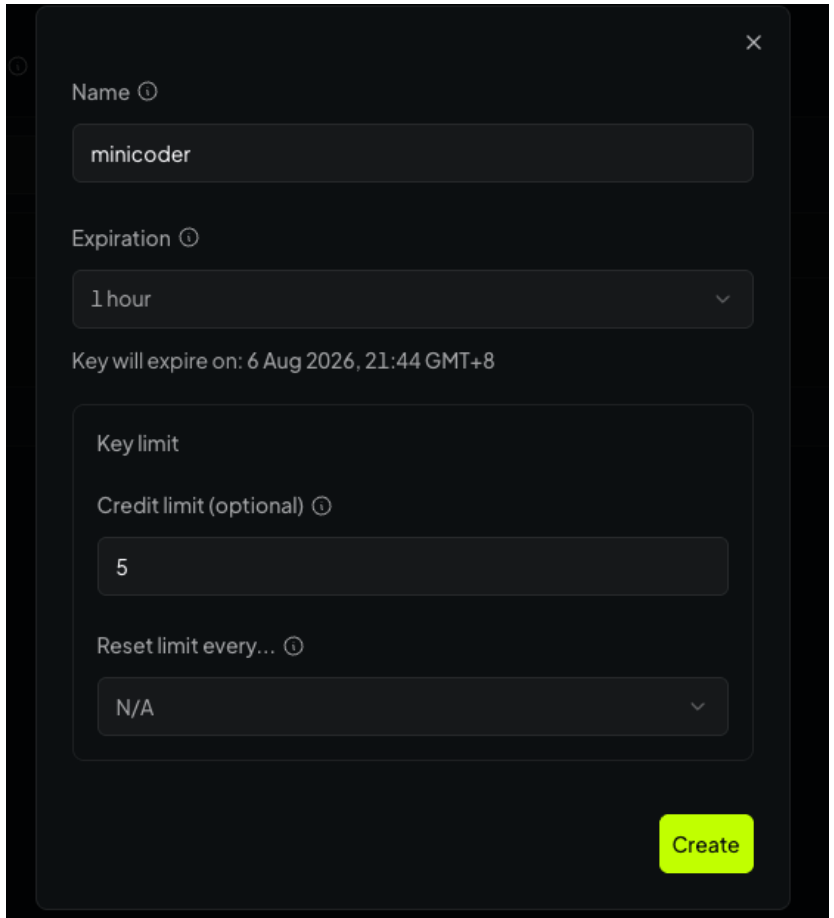
STEP02: calling a cloud LLM through API

- Next, we will need to create an API token.



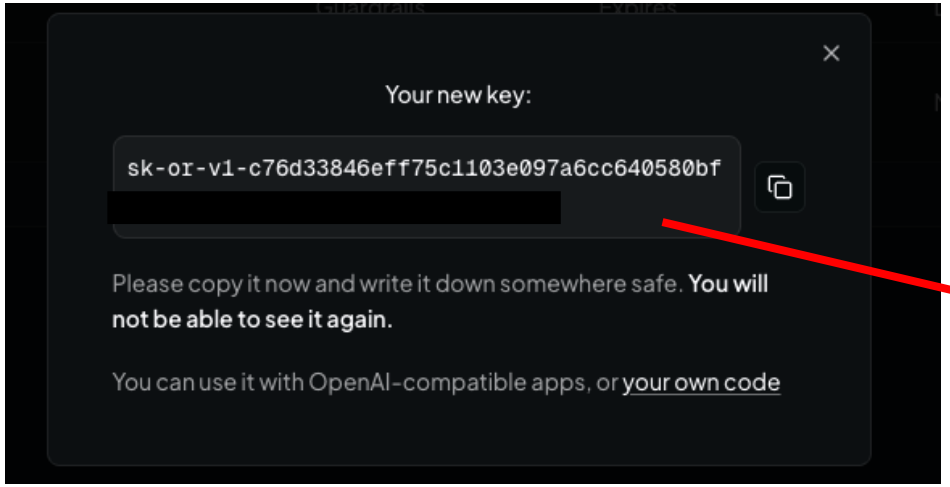
STEP02: calling a cloud LLM through API

- Set up the token as the following



STEP02: calling a cloud LLM through API

- You will see your token. Please copy-paste it into the .env file.
- The .env file should be generated in the project folder after you successfully run GitHub Copilot on SETUP.md.
- If you cannot find .env, it means either you didn't run or the run wasn't successful.




```
1  OLLAMA_HOST=http://localhost:11434
2  OLLAMA_API_KEY=
3  OLLAMA_MODEL=qwen3:0.6b
4
5  # Cloud backend used by milestone 03's multi-round session
6  # Get an API key at https://openrouter.ai/keys, then paste
7  # OPENROUTER_MODEL must be the exact model slug shown on h
8  LLM_BACKEND=ollama
9  OPENROUTER_API_KEY=
10 OPENROUTER_MODEL=
11
12
```

STEP02: calling a cloud LLM through API

- We will use DeepSeek V4 Flash
- Please put “~deepseek/deepseek-v4-flash-latest” in .env file.

```
1  OLLAMA_HOST=http://localhost:11434
2  OLLAMA_API_KEY=
3  OLLAMA_MODEL=qwen3:0.6b
4
5  # Cloud backend used by milestone 03's multi-round session
6  # Get an API key at https://openrouter.ai/keys, then paste
7  # OPENROUTER_MODEL must be the exact model slug shown on ht
8  LLM_BACKEND=ollama
9  OPENROUTER_API_KEY= sk-or-v1-c76d33846eff75c1103e097a6cc640580bf
10 OPENROUTER_MODEL=~deepseek/deepseek-v4-flash-latest
11
12
```

STEP02: calling a cloud LLM through API

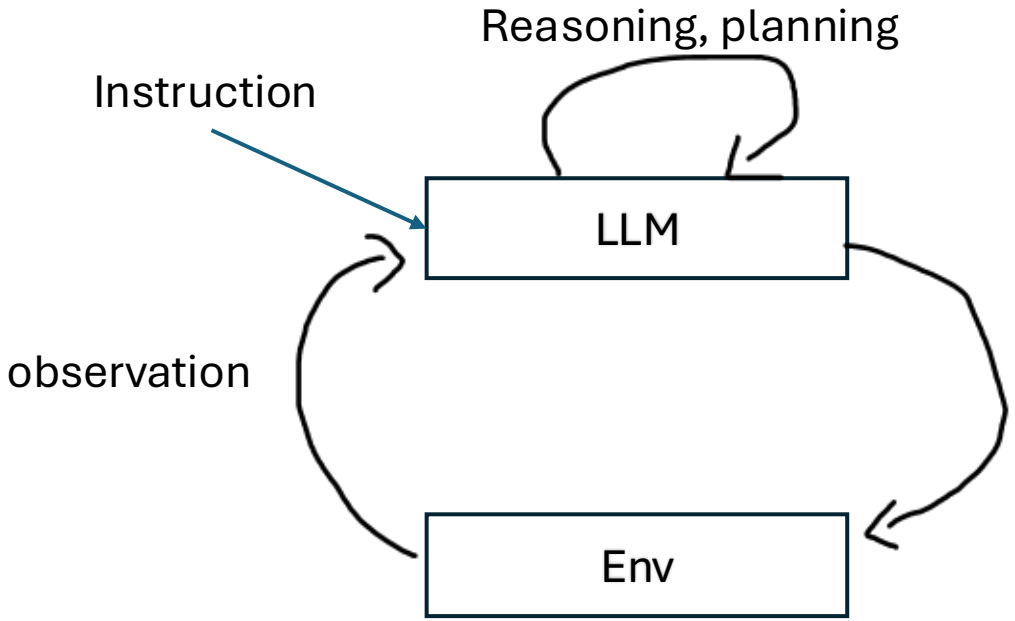
- Observe the output of DeepSeek V4 Flash (284B total parameters)
- Can it answer the question correctly?
- Model scale matters!

For modern, well-trained dense language models:

Model scale	Typical qualitative capability
Below 1B	Language fluency, memorized knowledge and simple instruction following; reasoning is fragile
1–3B	Basic generalization and elementary reasoning begin to become practically useful
7–14B	First range that often feels like a broadly capable general-purpose assistant
30–70B	More reliable reasoning, coding, knowledge integration and instruction following
100B+ active parameters	Historically associated with strong few-shot and chain-of-thought abilities, although newer training methods can obtain similar abilities at much smaller sizes
Frontier MoE models	Tens to roughly 100B active parameters, despite hundreds of billions or trillions of stored parameters

STEP03: build a mini coding agent with a cloud LLM

- A coding agent backed by LLM works in the following way




```
Available actions: list_files, search_text(query),
read_file(path), plan(steps),
verify(code), finish(summary).
```

STEP03: build a mini coding agent with a cloud LLM

- Let’s take a quick look at how the actions are defined as python code.
- LLM is supposed to reply in JSON format (key-value pair), which is a common practice for agentic LLMs. It’s easier to manage the response.

STEP03: build a mini coding agent with a cloud LLM

- Then we look at the coding question a mini coding agent is trying to solve.

STEP03: build a mini coding agent with a cloud LLM

- After the agent is done, we take a look at the interaction between the LLM and the environment (in this case, our laptop)
- How many steps did the agent take?
- What is the purpose of each step?
- How did we make sure the final output of LLM is reliable?

STEP03: build a mini coding agent with a cloud LLM

- We have a test suite to support the action “verify”!
- If the LLM fails in the first attempt when calling verify, it will receive the error message and reason by itself and plan new actions and try again.
- It will iterate over this loop until it passes the “verify” step.
- What does above mentioned process tell us?

STEP03: build a mini coding agent with a cloud LLM

- **We Need to Redefine Debugging in the Era of Agentic AI**
- **Traditional Debugging:** Read the error message. -> Diagnose the root cause. -> Modify the code. -> Run the program again. -> Repeat until it works.
- **Today, coding agents can automate much of this entire loop.**

STEP03: build a mini coding agent with a cloud LLM

- **So What Does "Debugging" Mean Now?**
- The engineer's role shifts from **fixing code** to **providing verifiable evidence** that the code is correct.
- Instead of spending most of our time writing fixes, we increasingly spend our time defining how the solution should be verified, for example by:
  - Writing comprehensive test suites.
  - Specifying acceptance criteria.
  - Designing edge cases and failure scenarios.
- **In the age of coding agents, debugging is no longer just fixing bugs—it is designing the verification process that allows AI to find and fix bugs correctly.**

STEP03: build a mini coding agent with a cloud LLM

- What if AI can provide its own verifiable mechanism?
- …
- There are many more interesting questions waiting for you to explore.

Wrapping up

- We learn how to
- Use GitHub Copilot to approach a tiny project
- Run a local LLM on laptop
- Run a cloud LLM via API
- Build a mini coding agent backed by cloud LLM.